



NORMANHURST BOYS' HIGH SCHOOL
NEW SOUTH WALES

Student Number

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Class: 12M 1 2 3 4 5 (Please Circle)

2013

**HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION**

Mathematics Extension 1

General Instructions

- Reading time - 5 minutes
- Working time - 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11-14, show relevant mathematical reasoning and/or calculations
- Begin each question on a separate writing booklet

Total marks - 70

Section I

Pages 2-5

10 marks

- Attempt Questions 1-10
- Answer on the Multiple Choice answer sheet provided
- Allow about 15 minutes for this section

Section II

Pages 6-11

60 marks

- Attempt Questions 11-14
- Allow about 1 hour 45 minutes for this section

Students are advised that this is a school-based examination only and cannot in any way guarantee the content or format of future Higher School Certificate Examinations.

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10

- (1) The height of a giraffe has been modelled using the equation:

$$H = 5.40 - 4.80e^{-kt}$$

where H is the height in metres, t is the age in years and k is a positive constant.

If a 6 years old giraffe has a height of 5.16 metres, find the value of k , correct to 2 significant figures.

- (A) 0.05
- (B) 0.24
- (C) 0.50
- (D) 4.8

- (2) What is the value of $\lim_{x \rightarrow 0} \left(\frac{\sin \frac{1}{3}x}{2x} \right)$

- (A) $\frac{1}{6}$
- (B) $\frac{2}{3}$
- (C) $\frac{3}{2}$
- (D) 6

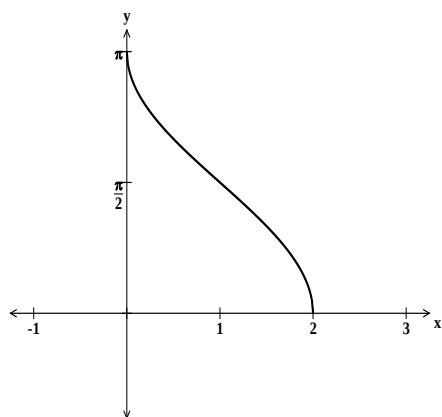
- (3) Which of the following equates to the expression $\frac{1 - e^{3x}}{1 - e^{2x}}$.

- (A) $1 + \frac{e^{2x}}{1 + e^x}$
- (B) $1 - e^x$
- (C) $1 + e^x + e^{2x}$
- (D) None of the above

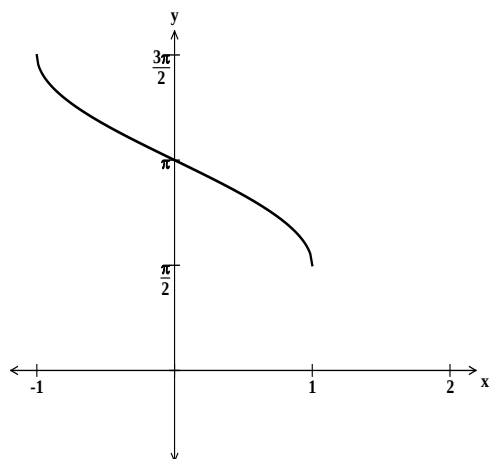
- (4) The point P divides the interval $A(\frac{17}{3}, 2)$ to $B(-3, 4)$ **externally** in the ratio $2:3$. Which one of the following is the coordinates of point P ?
- (A) $(-23, 2)$
- (B) $(-9, -12)$
- (C) $(9, 0)$
- (D) $(23, -2)$
- (5) A curve is defined by the parametric equations $x = \sin 2t$ and $y = \cos 2t$. Which of the following, in terms of t , equates to $\frac{dy}{dx}$?
- (A) $-\tan 2t$
- (B) $2 \tan 2t$
- (C) $2 \sin 4t$
- (D) $\cos 4t$
- (6) Which of the following is the inverse function of $y = \frac{x-4}{x-2}$, $x \neq 2$?
- (A) $y = \frac{x-2}{x-4}$
- (B) $y = f^{-1}(y)$
- (C) $y = \frac{2(x-2)}{x-1}$
- (D) $y = \frac{x+4}{x+2}$

(7) Which of the following represents the graph of $y = \cos^{-1}(x+1)$.

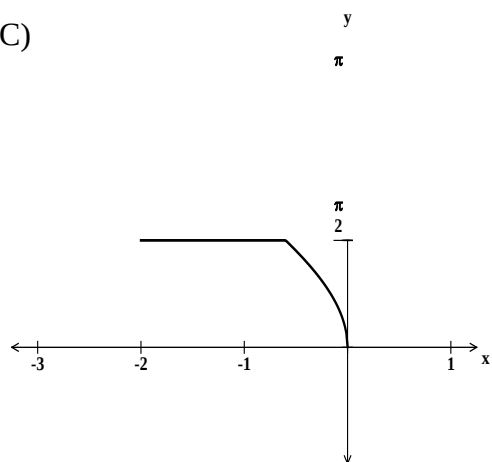
(A)



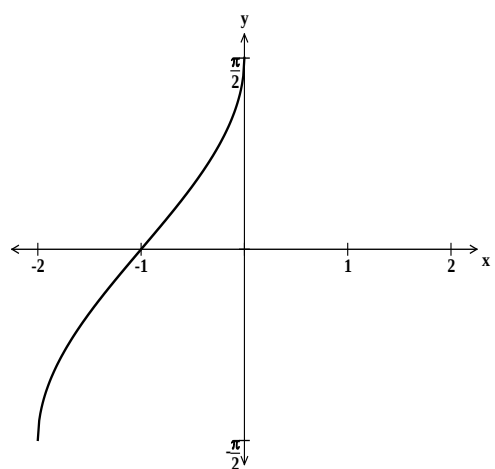
(B)



(C)



(D)



(8) Given that $xy = x+1$, the definite integral $\int_3^5 x \, dy$ equates to:

(A) e^2

(B) $\ln 2$

(C) $-\ln\left(\frac{5}{3}\right)$

(D) $e^{\frac{5}{3}}$

- (9) The motion of a particle moving along the x – axis executes simple harmonic motion. The maximum velocity of the particle is 4 m/s and the period of motion is π seconds . Which of the following could be the displacement equation for this particle?

- (A) $x = 4 \cos \pi t$
- (B) $x = -\sin 2t$
- (C) $x = 2 \cos 2t$
- (D) $x = 2 + \cos 2t$

- (10) A particle moves with a velocity $v \text{ m/s}$ where $v = \sqrt{x^2 + 1}$. Given that $x > 0$, which of the following is equal to the acceleration of the particle when $v = 4 \text{ m/s}$.

- (A) $\sqrt{17} \text{ m/s}^2$
- (B) -3 m/s^2
- (C) $\sqrt{15} \text{ m/s}^2$
- (D) $2\sqrt{17} \text{ m/s}^2$

Section II

60 marks

Attempt Questions 11-14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11-14, your responses should include relevant mathematical reasoning and/or calculations.

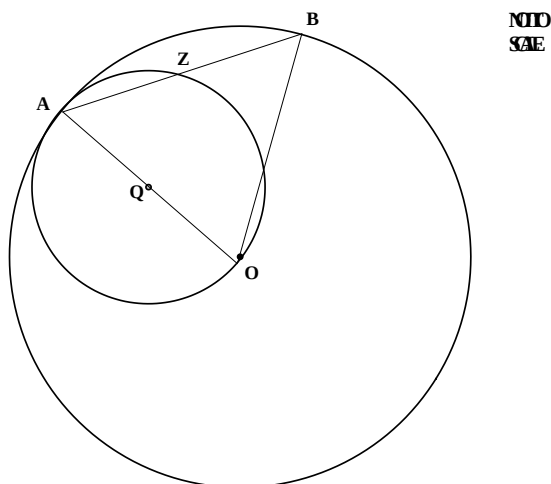
Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) Find the exact value of $\sin 75^\circ$. 2
- (b) Given that the acute angle between the lines $y = mx$ and $2x - 3y = 0$ is 45° , find possible value(s) of m . 3
- (c) Using the substitution $u = 1 + x^2$, or otherwise, evaluate 3
- $$\int_0^{\sqrt{8}} \left(\frac{x}{\sqrt{1+x^2}} \right) dx.$$
- (d) Solve the following inequality for x : 3
- $$\frac{1}{x} + \frac{x}{(x-2)} < 0$$
- (e) (i) On the same number plane, graph the following functions: 2
- $$y = 4 - x^2 \quad \text{and} \quad y = |3x|$$
- (ii) Hence or otherwise solve $4 - x^2 \leq |3x|$ 2

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) Find the exact value of $\sin\left(2\cos^{-1}\frac{\sqrt{3}}{4}\right)$. 2

- (b) AB is a chord of a circle centre O . AO is a diameter of a circle centre Q . Z is the point where the circle centre Q meets AB .



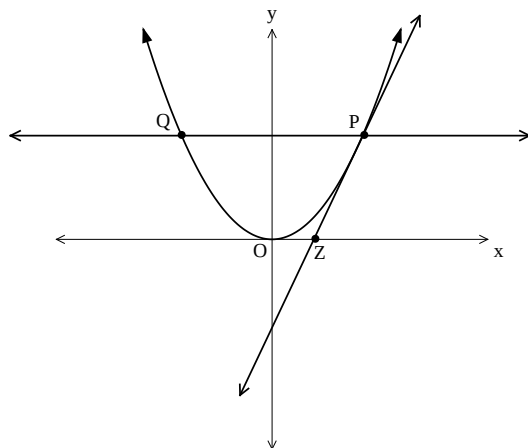
**NOT
SAE**

- (i) Explain why $AO = OB$. 1
- (ii) Hence or otherwise, prove that $AZ = ZB$. 2
- (c) The quadratic equation $x^2 - 4x + 9 = 0$ has roots $\tan A$ and $\tan B$. Hence, find the size(s) of $\angle(A+B)$, noting that $0 \leq A+B \leq 360^\circ$ (leave your answer to the nearest degree). 3
- (d) (i) By use of long division, find the remainder, in terms of a and b when $P(x) = x^4 + 3x^3 + 6x^2 + ax + b$ is divided by $x^2 + 2x + 1$. 2
- (ii) If this remainder is $3x + 2$, find the values of a and b . 1
- (e) (i) Prove that $\sin A \cos A \cos 2A = \frac{1}{4} \sin 4A$. 2
- (ii) Hence or otherwise solve $\sin A \cos A \cos 2A = 0$, for $0 \leq A \leq \frac{\pi}{2}$. 2

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) Prove by mathematical induction that $5^n \geq 1 + 4n$ for all integers $n \geq 1$. 3

(b)



$P(2ap, ap^2)$ and $Q(-2ap, ap^2)$ are variable points on the parabola $x^2 = 4ay$. The line PQ is parallel to the x -axis. The tangent at P meets the x -axis at Z .

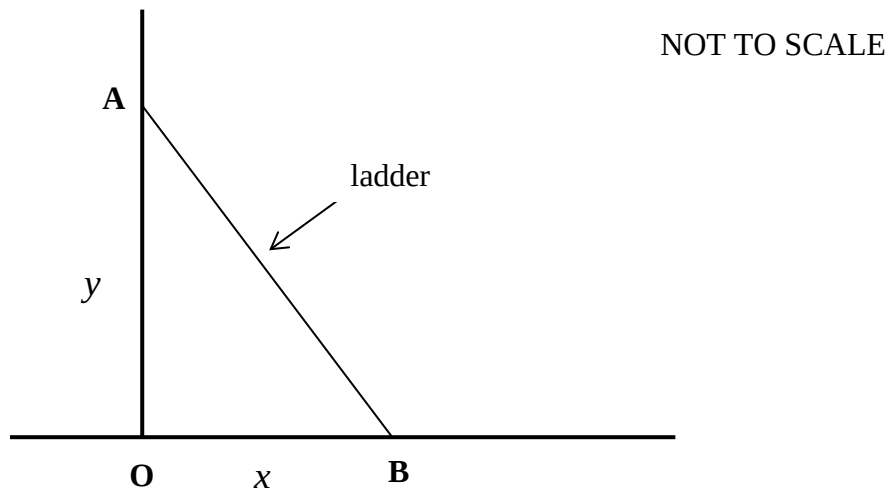
- (i) Show that the equation of the tangent at P is given by $y = px - ap^2$ 2
 - (ii) Hence show that $Z = (ap, 0)$. 1
 - (iii) Find the locus of midpoints of QZ . 2
- (c)
- (i) Graph the function $y = 2 \tan^{-1}(x)$. 1
 - (ii) Graphically show why $2 \tan^{-1}(x) - \frac{x}{4} = 0$ has one root, for $x > 0$. 1
 - (iii) Taking $x_1 = 10$ as a first approximation to this root, use one application of Newton's method to find a better approximation, correct to 2 decimal places. 2

Question 13 continues on page 9

Question 13 (continued)

13 (d)

3



A ladder AB, 5 metres long, is leaning against a vertical wall OA, with its foot B, on horizontal ground OB. The distances OB and OA are x and y metres respectively. x and y are related by the equation $x^2 + y^2 = 25$.

The foot of the ladder begins to slide along the ground away from the wall at a constant speed of 1 metre per second.

Find the speed at which the top of the ladder A is moving down the wall at the time when the top of the ladder is 4 metres above the ground.

End of Question 13

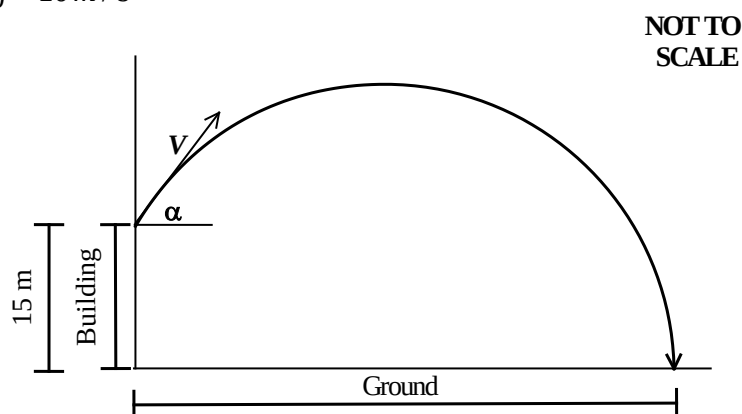
Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) A particle is travelling in a straight line. Its displacement (x cm) from O at a given time (t sec) after the start of motion is given by: $x = 2 + \sin^2 t$.
- (i) Prove that the particle is undergoing simple harmonic motion. 2
- (ii) Find the centre of motion. 1
- (iii) Find the total distance travelled by the particle in the first $\frac{3\pi}{2}$ seconds. 2
- (b) A shade sail with corners A , B and C is shown in diagram 1, supported by three vertical posts. The posts at corners A and C are the same height, and the post at corner B is 2.4 m taller. Diagram 2 shows the sail in more detail. D is the point on the taller post horizontally level with the tops of the other two posts. $AD = 6.4$ m and $DC = 5.2$ m. $\angle ADC = 125^\circ$. 3
Find the area of the shade sail ABC (leave your answer to 1 decimal place)

Question 14 continues on page 11

Question 14 (continued)

- (c) Over 80 years ago, during training exercises, the Army fired an experimental missile from the top of a building 15 m high with initial velocity (v) where $v = 130 \text{ m/s}$, at an angle (α) to the horizontal. Noting that $\alpha = \tan^{-1}\left(\frac{5}{12}\right)$ and taking $g = 10 \text{ m/s}^2$



Assume that the equations of motion of the missile are $\ddot{x} = 0$ and $\ddot{y} = -10$

- | | | |
|-------|---|---|
| (i) | Show that $\dot{x} = 120$ and $\dot{y} = -10t + 50$.
Hence write down the equations of x and y . | 2 |
| (ii) | The rocket hit its intended target when its velocity reached $60\sqrt{5} \text{ m/s}$.
Find the horizontal distance that the missile travelled to hit its target. | 2 |
| (iii) | The rocket was designed to hit its target once the angle to the horizontal of its flight path in a downward direction lies between 20° and 30° . Find the range of times after firing that this could happen. | 3 |

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Normanhurst Boys High School
2013 HSC TRIAL EXAMINATION

MATHEMATICS EXTENSION 1 – MARKING GUIDELINES

Section I

Question	Marks	Answer	Outcomes Assessed
1	1	XC	O1
2	1	A	O1
3	1	A	O1
4	1	D	O3
5	1	A	O4
6	1	C	O4
7	1	C	O4
8	1	B	O5
9	1	C	O5
10	1	C	O5

Question 11 (15 marks)

11(a) (2 marks)

Outcomes Assessed: O3

Criteria	Marks
• Use sine of the sum	1
• Correct answer	1

Answer

$$\begin{aligned}\sin 75^\circ &= \sin(45^\circ + 30^\circ) \\ &= \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ \\ &= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \\ &= \frac{\sqrt{3} + 1}{2\sqrt{2}}\end{aligned}$$

11(b) (3 marks)

Outcomes Assessed: O3

Criteria	Marks
• Obtains correct gradients	1
• Correct substitution into formulae	1
• Correct answer	1

Answer

$$\begin{aligned}\tan 45^\circ &= \frac{m - \frac{2}{3}}{1 + \frac{2m}{3}} \\ \frac{3m - 2}{3 + 2m} &= 1 \\ \frac{3m - 2}{3 + 2m} &= 1 \quad \text{or} \quad \frac{3m - 2}{3 + 2m} = -1 \\ m = 5 \quad \text{or} \quad m &= -\frac{1}{5}\end{aligned}$$

11(c) (3 marks)

Outcomes Assessed: O5

Criteria	Marks
• Obtains correct limits	1
• Obtains $I = \frac{1}{2} \int_1^9 \left(\frac{du}{u^{\frac{1}{2}}} \right)$	1
• Correct answer	1

Answer

$$\begin{aligned}u &= 1 + x^2 \\ \frac{1}{2} du &= x dx \\ x = \sqrt{8} &\rightarrow u = 9 \\ x = 0 &\rightarrow u = 1\end{aligned}$$

$$\begin{aligned}I &= \int_0^{\sqrt{8}} \left(\frac{x}{\sqrt{1+x^2}} \right) dx \\ I &= \frac{1}{2} \int_1^9 \left(\frac{du}{u^{\frac{1}{2}}} \right) \\ &= \left[u^{\frac{1}{2}} \right]_1^9 \\ &= 3 - 1 \\ &= 2\end{aligned}$$

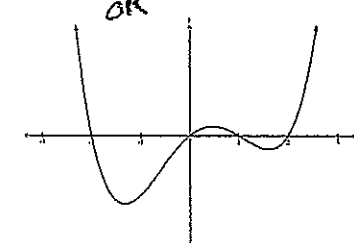
11(d) (3 marks)

Outcomes Assessed: O2

Criteria	Marks
• Multiplies throughout by $x^2(x-2)^2$	1
• Obtain $x(x-2)(x+2)(x-1) < 0$	1
• Correct answer	1

Answer

$$\begin{aligned}\frac{1}{x} + \frac{x}{(x-2)} &< 0 \\ x(x-2)^2 + x^2(x-2) &< 0 \\ x(x-2)[(x-2) + x^2] &< 0 \\ x(x-2)[x^2 + x - 2] &< 0 \\ x(x-2)(x+2)(x-1) &< 0 \\ -2 < x < 0 \quad \text{or} \quad 1 < x < 2 \quad (\text{from diagram})\end{aligned}$$

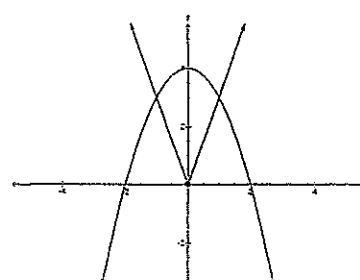


11(e) (i) (2 marks)

Outcomes Assessed: O1

Criteria	Marks
• Correct graph for $y = 4 - x^2$	1
• Correct graph for $y = 3x $	1

Answer



11(e) (ii) (2 marks)

Outcomes Assessed: O1

Criteria	Mark
• Solve for both points of intersection	1
• Correct answer	1

Answer

Solve

$$4 - x^2 = 3x$$

$$4 - x^2 = 3x \quad \text{or} \quad 4 - x^2 = -3x$$

$$x^2 + 3x - 4 = 0 \quad \text{or} \quad x^2 - 3x - 4 = 0$$

$$(x+4)(x-1) = 0 \quad \text{or} \quad (x-4)(x+1) = 0$$

$$x = 1, -4 \quad x = 4, -1$$

check solutions

correct solutions: $x = \pm 1$

hence from diagram $x < -1$ or $x > 1$

Question 12 (15 marks)

12(a) (2 marks)

Outcomes Assessed: O4

Criteria	Marks
• Achieves $\cos \alpha = \frac{\sqrt{3}}{4}$ and $\sin \alpha = \frac{\sqrt{13}}{4}$	1
• Correct answer	1

Answer

$$\text{Let } \alpha = \cos^{-1} \frac{\sqrt{3}}{4}$$

$$\therefore \sin \left(2 \cos^{-1} \frac{\sqrt{3}}{4} \right) = \sin(2\alpha)$$

$$\text{Also } \cos \alpha = \frac{\sqrt{3}}{4}, \text{ hence } \sin \alpha = \frac{\sqrt{13}}{4} \text{ (pythagoras)}$$

$$\sin \left(2 \cos^{-1} \frac{\sqrt{3}}{4} \right) = \sin(2\alpha)$$

$$= 2 \sin \alpha \cos \alpha$$

$$= 2 \cdot \frac{\sqrt{13}}{4} \cdot \frac{\sqrt{3}}{4}$$

$$= \frac{\sqrt{39}}{8}$$

12(b) (i) (1 mark)

Outcomes Assessed: O3

Criteria	Mark
• Correct answer reasoning	1

Answer

$$AO = OB \quad (\text{radii of circle centre } O)$$

12(b) (ii) (2 marks)

Outcomes Assessed: O3

Criteria	Marks
• Notes that $\angle AZO = 90^\circ$ with reasons	1
• Correct answer with correct reasoning	1

Answer

If from (i) $\triangle OAB$ is isosceles

Also $\angle AZO = 90^\circ$ (angles in a semi-circle are right angles at the circumference)

$\therefore AZ = ZB$ (a line from the apex of an isosceles triangle which meets the base at right angles, bisects the base)

12(c) (3 marks)

Outcomes Assessed: O3

Criteria	Marks
• Obtains $\tan A + \tan B = 4$ or $\tan A \tan B = 9$	1
• Uses $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$	1
• Correct answer	1

Answer

$$\tan A + \tan B = -\frac{b}{a}$$

$$\tan A + \tan B = 4$$

$$\tan A \tan B = \frac{c}{a}$$

$$\tan A \tan B = 9$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$= \frac{4}{1-9}$$

$$= -\frac{1}{2}$$

$$\angle(A+B) = 153^\circ 26', 333^\circ 26'$$

$$= 153^\circ, 333^\circ$$

12(d) (i) (2 marks)

Outcomes Assessed: O4

Criteria	Marks
• Makes a positive attempt to solve obtain the remainder by long division	1
• Correct answer	1

Answer

$$P(x) = x^4 + 3x^3 + 6x^2 + ax + b$$

By long division

$$P(x) = (x^2 + 2x + 1)(x^2 + x + 3) + [(a-7)x + (b-3)]$$

$$\therefore R(x) = (a-7)x + (b-3)$$

12(d) (ii) (1 mark)

Outcomes Assessed: O4

Criteria	Mark
• Correct answer	1

Answer

$$3x + 2 = (a-7)x + (b-3)$$

$$\therefore a = 10 \text{ and } b = 5$$

12(e) (i) (2 marks)

Outcomes Assessed: O3

Criteria	Marks
• Uses the double angle correctly at least once	1
• Correct answer with correct working	1

Answer

$$\sin A \cos A \cos 2A = \frac{1}{4} \sin 4A$$

$$LHS = \frac{1}{2} [2 \sin A \cos A \times \cos 2A]$$

$$= \frac{1}{2} [\sin 2A \times \cos 2A]$$

$$= \frac{1}{2} \times \frac{1}{2} [2 \sin 2A \times \cos 2A]$$

$$= \frac{1}{4} \sin 4A$$

$$= RHS$$

12(e) (ii) (2 marks)

Outcomes Assessed: O3

Criteria	Marks
• Achieves at least 2 of the correct answers	1
• Correct answer	1

Answer

$$\frac{1}{4} \sin 4A = 0$$

$$\sin 4A = 0$$

$$4A = 0, \pi, 2\pi, 3\pi, 4\pi, \dots$$

$$A = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \dots$$

$$\text{Hence } A = 0, \frac{\pi}{4}, \frac{\pi}{2} \quad ; 0 \leq A \leq \frac{\pi}{2}$$

Question 13 (15 marks)

13(a) (3 marks)

Outcomes Assessed: O2

Criteria	Marks
• Provides clear steps similar to the first three steps below	1
• Makes a substitution in step 3 (from step 2)	1
• Provides a clear working to obtain the required result and provides a conclusion	1

Answer

$$5^n \geq 1 + 4n$$

Step 1: Prove the expression is true for $n=1$

$$5 \geq 5 \quad (\text{true}) \quad LHS = 5^1 = 5 \quad RHS = 1 + 4 \times 1 = 5$$

Assume the expression is true for $n=k$ (where k is even)

$$i.e. 5^k \geq 1 + 4k \quad (\text{where } k \text{ is a positive integer})$$

Step 2: Prove the expression is true for $n=k+1$

$$i.e. 5^{k+1} \geq 1 + 4(k+1)$$

$$5 \cdot 5^k \geq 4k + 5$$

$$5^{k+1} - 4k - 5 \geq 0$$

$$\text{Now } LHS = 5 \cdot 5^k - 4k - 5$$

$$\geq 5(1 + 4k) - 4k - 5 \quad (\text{from assumption})$$

$$= 16k \geq 0 \quad (k \geq 1)$$

Hence if the expression is true when $n=k$, it is true when $n=k+1$

If the expression is true for $n=1$, $\therefore$ it is true when $n=2$

If true for $n=2$, $\therefore$ it is true when $n=3$

Therefore the expression is true for all $n, n \geq 1$.

Step 3

Step 1 & 3

Step 2 - Using assumption for $n=k$

- All correct

13(b) (i) (2 marks)

Outcomes Assessed: O4

Criteria	Marks
• Show gradient of tangent at $P = p$	1
• Achieves $y = px - ap^2$ with sufficient working	1

Answer $x^2 = 4ay$

$$\frac{dy}{dx} = \frac{x}{2a} = \frac{2ap}{2a} = p$$

Gradient of tangent at $P = \frac{2ap}{2a} = p$

Equation of tangent at P

$$y - ap^2 = p(x - 2ap)$$

$$y - ap^2 = px - 2ap^2$$

$$y = px - ap^2$$

(2)

13(b) (ii) (1 mark)

Outcomes Assessed: O4

Criteria	Marks
• Correct working to achieve Z	1

Answer

Tangent: $y = px - ap^2$

For Z: substitute $y=0$

$$px - ap^2 = 0$$

$$x = ap$$

$$Z = (ap, 0)$$

(1)

13(b) (ii) (2 marks)

Outcomes Assessed: O4

Criteria	Marks
• Obtains midpoint $z_2 = \left(\frac{-ap}{2}, \frac{ap^2}{2}\right)$	1
• Correct answer for locus	1

Answer

$Q = (-2ap, ap^2)$ symmetry of parabola

$\therefore$ midpoint $z_2 = \left(\frac{-ap}{2}, \frac{ap^2}{2}\right)$

hence $x = \frac{-ap}{2}$

$p = \frac{-2x}{a}$

Sub into ②,

$y = \frac{ap^2}{2}$

$y = \frac{a}{2} \left(\frac{-2x}{a}\right)^2$

$y = \frac{2x^2}{a}$

$\therefore 2x^2 = ay$ is the locus of midpoints of QZ.

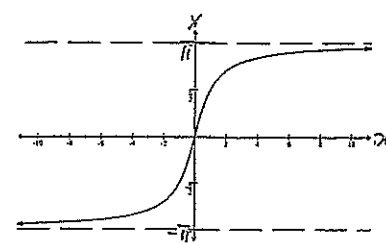
(2)

13(c) (i) (1 mark)

Outcomes Assessed: O4

Criteria	Mark
• Correct diagram graph with asymptotes labeled.	1

Answer



(1)

13(c) (ii) (1 mark)

Outcomes Assessed: O4

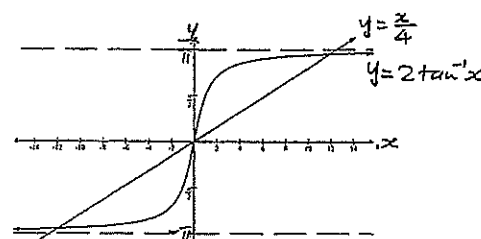
Criteria	Mark
• Correct answer	1

Answer

$$2 \tan^{-1}(x) - \frac{x}{4} = 0$$

$$2 \tan^{-1}(x) = \frac{x}{4}$$

$\therefore$ Graph $y = 2 \tan^{-1}(x)$ and $y = \frac{x}{4}$ to show that there is only one point of intersection $(2, 3)$ for $x > 0$



(1)

13(c) (iii) (2 marks)

Outcomes Assessed: O4

Criteria	Marks
• Uses the correct formulae and makes a good attempt at achieving answer	1
• Correct answer	1

Answer

$$P(x) = 2 \tan^{-1}(x) - \frac{x}{4}$$

$$P'(x) = \frac{2}{1+x^2} - \frac{1}{4}$$

$$x_2 = x_1 - \frac{P(x_1)}{P'(x_1)}$$

$$= 10 - \frac{P(10)}{P'(10)}$$

$$= 10 + \frac{0.4423}{0.230}$$

$$= 11.92 \text{ (2 decimal places)}$$

(2)

13(d) (3 marks)

Outcomes Assessed: O5

Criteria	Marks
• Differentiates correctly	1
• Equates various rates correctly	1
• Achieves correct answer	1

Answer

$$x^2 + y^2 = 25$$

$$y = \sqrt{25 - x^2} \quad (y > 0)$$

$$\frac{dy}{dx} = \frac{-x}{\sqrt{25 - x^2}}$$

Now $\frac{dx}{dt} = 1$

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

$$\frac{dy}{dt} = \frac{-x}{\sqrt{25 - x^2}} \times 1$$

Now when $y = 4$, $x = 3$

$$\frac{dy}{dt} = \frac{-3}{\sqrt{25 - 3^2}} \times 1$$

$$= -\frac{3}{4} \text{ (i.e. } \frac{3}{4} \text{ metres per second down the wall)}$$

(2)

Note: A quicker method using the implicit function rule can be used by Ext 2 students.

Question 14 (15 marks)

14(a) (i) (2 marks)

Outcomes Assessed: O5

Criteria	Marks
• Achieves the acceleration formulae in any form	1
• Correct working	1

Answer

$$x = 2 + \sin^2 t \quad \text{--- ①}$$

$$\dot{x} = 2 \sin t \cos t$$

$$\ddot{x} = \cos t (2 \cos t) + 2 \sin t (-\sin t) \quad \checkmark$$

$$= 2 [\cos^2 t - \sin^2 t]$$

$$= 2 [1 - \sin^2 t - \sin^2 t]$$

$$= 2 [1 - 2 \sin^2 t]$$

$$= 2 [1 - 2(x - 2)] \text{ using ①}$$

$$= 2 [1 - 2x + 4]$$

$$= 10 - 4x$$

$$= -4 \left(x - \frac{5}{2}\right) \quad \checkmark$$

Since in the form of $\ddot{x} = -n^2 X$, therefore SHM.

where $X = (x - \frac{5}{2})$ and $n = 2$

(2)

14(a) (ii) (1 mark)

Outcomes Assessed: O5

Criteria	Mark
• Correct answer	1

Answer

$$\ddot{x} = 0$$

$$10 - 4x = 0$$

$$x = \frac{5}{2} \quad \checkmark$$

(1)

14(a) (iii) (2 marks)

Outcomes Assessed: O5

Criteria	Marks
Achieves $t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, \dots$	1
Correct answer with sufficient working	1

Answer

$v = 0 \leftarrow$ Particle changes direction

$$v = 2 \sin t \cos t$$

$$2 \sin t \cos t = 0$$

$$\therefore \sin t = 0$$

$$\cos t = 0$$

$$t = 0, \pi, 2\pi, 3\pi, \dots \quad t = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$$

$$t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, \dots$$

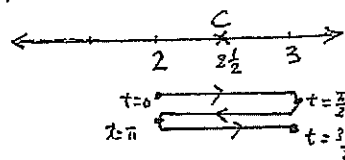
when $t = 0$ $x = 2$

$$t = \frac{\pi}{2} \quad x = 3$$

$$t = \pi \quad x = 2$$

$$t = \frac{3\pi}{2} \quad x = 3$$

$$\therefore \text{total distance} = 3 \text{ cm}$$



(2)

14(b) (3 marks)

Outcomes Assessed: O3

Criteria	Mark
Find AC	1
Find $\angle ABC$	1
Correct answer to area	1

Answer

$$\triangle ABD, \quad AB^2 = 2.4^2 + 6.4^2$$

$$AB \approx 6.835 \text{ m}$$

$$\triangle CBD, \quad BC^2 = 2.4^2 + 5.2^2$$

$$BC \approx 5.727 \text{ m}$$

$$\triangle ACD, \quad AC^2 = 6.4^2 + 5.2^2 - 2 \times 6.4 \times 5.2 \times \cos 125^\circ$$

$$AC \approx 10.304 \text{ m}$$

$$\triangle ABC, \quad \cos \angle ABC = \frac{AB^2 + BC^2 - AC^2}{2(AB)(BC)}$$

$$\angle ABC \approx 109.9^\circ$$

$$\text{Area } \triangle ABC = \frac{1}{2} \times 6.835 \times 5.727 \times \sin 109.9^\circ$$

$$= 18.4 \text{ m}^2 \text{ (1dp)}$$

(3)

14(c) (i) (2 marks)

Outcomes Assessed: O5

Criteria	Marks
Show $\dot{x}$ and $\dot{y}$ with sufficient working	1
Correct x and y with working shown	1

Answer

$$\tan \alpha = \frac{5}{12}, \therefore \sin \alpha = \frac{5}{13} \text{ and } \cos \alpha = \frac{12}{13}$$

$$\ddot{x} = 0$$

$$\ddot{y} = -10$$

$$\dot{x} = v \cos \alpha \quad \dot{y} = -10t + v \sin \alpha$$

$$x = 130 \times \frac{12}{13} \quad y = -10t + 130 \times \frac{5}{13}$$

$$x = 120 \quad y = -10t + 50$$

$$x = 120t \quad y = -5t^2 + 50t + 15$$

$$\ddot{x} = 0$$

$$\ddot{y} = -10$$

$$\text{when } t = 0, \dot{x} = v \cos \alpha$$

$$\therefore C_1 = v \cos \alpha$$

$$\dot{x} = v \cos \alpha$$

$$= 130 \times \frac{12}{13}$$

$$= 120$$

$$x = 120t + C_1$$

$$\text{when } t = 0, x = 0$$

$$\therefore C_1 = 0$$

$$x = 120t$$

$$\ddot{y} = -10$$

$$\text{when } t = 0, \dot{y} = v \sin \alpha$$

$$\therefore C_2 = v \sin \alpha$$

$$\dot{y} = -10t + v \sin \alpha$$

$$= -10t + 130 \times \frac{5}{13}$$

$$= -10t + 50$$

$$y = -\frac{10t^2}{2} + 50t + C_2$$

$$\text{when } t = 0, y = 15$$

$$\therefore 15 = C_2$$

$$y = -5t^2 + 50t + 15$$

14(c) (ii) (2 marks)

Outcomes Assessed: O5

Criteria	Marks
Uses $v^2 = \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2$ and works towards answer	1
Correct answer	1

Answer

$$v^2 = \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2$$

$$(60\sqrt{5})^2 = (120)^2 + (-10t + 50)^2$$

$$18000 = 14400 + 100t^2 - 1000t + 2500$$

$$100t^2 - 1000t - 1100 = 0$$

$$t^2 - 10t - 11 = 0$$

$$(t-11)(t+1) = 0$$

$$t = 11, -1$$

$$t = 11 \quad (t > 0)$$

$$x = 120(11)$$

$$= 1320 \text{ m}$$

(2)

14(c) (iii) (3 marks)
Outcomes Assessed: O5

Criteria	Marks
Uses $\tan(20^\circ) \leq \frac{y}{x} \leq \tan(30^\circ)$ or similar	1
Flight path is in a downward direction $\therefore$ negative	1
Correct answer	1

Answer

$$20^\circ < \theta < 30^\circ$$

$$\tan(20^\circ) < \frac{y}{x} < \tan(30^\circ)$$

$$0.364 < \frac{-10t + 50}{120} < 0.577$$

$$43.68 < -10t + 50 < 69.28$$

As the flight path is in a downward direction $y < 0 \Rightarrow 43.68 < 10t - 50 < 69.28$

Hence

$$50 - 10t < -43.68 \quad 50 - 10t > -69.28$$

$$9.868 \leq t \quad 11.928 \geq t$$

$$9.868 < t < 11.928$$

$$9.87 < t < 11.93$$

(3)

END OF PAPER